

Richard Prange

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EDUCATION

Western Carolina University

Bachelor of Science in Computer Science

Cullowhee, NC

May, 2025

Western Carolina University

Bachelor of Science in Mathematics

Cullowhee, NC

May, 2025

Relevant Courses: Data Structures and Algorithms, Software Development & Engineering, Database Systems, Cloud Computing, Information Security, Computer Networking

EXPERIENCE

Mathematics and Computer Science Tutor

Western Carolina University

Cullowhee, NC

January, 2022 → May, 2025

- Provide personalized assistance for drop in students
- Foster critical thinking and problem solving skills through active learning techniques
- Helped students learn and retain abstract mathematics and computer science topics, leading to an increase in re-enrollment rates

Junior Software Engineer

ally Financial

Charlotte, NC

February, 2026 → Current

- Worked in an agile environment, designing and implementing software and cloud solutions for the Deposits team
- Designed and implemented secure and robust RESTful and GraphQL apis

PROJECTS

WheeBFFit | *React, TypeScript, Node.js, Express, Springboot, JWT, Heroku, MySQL*

May, 2025

- Developed and deployed a full stack web application for a client
- Implemented a highly interactive student interface designed to keep students entertained along with a dashboard which allows teachers to track, display, and monitor a class' progress
- Information is stored in a SQL database and is accessible to authorized users
- **Available at:** <https://wheebffit.com>

ChessMeister | *Java, JavaFX*

May, 2024

- Developed a chess program that allows 2-4 players to play chess through a CLI or GUI
- Implemented valid check and checkmate algorithms, used to end the game and validate moves
- Incorporated functionality to allow for users to undo and redo moves
- Designed a loosely coupled system using interfaces and model view controller to allow for seamless future expansion on the user interface
- **Available at:** <https://gitfront.io/r/ricwp12/qsDUvfM56CWG/ChessMeisterV2/>

Rusty Battleship | *Rust*

January, 2025

- Developed a program that allows N-players to connect to a server and play a CLI version of BattleShip
- Provided functionality using threads that allow users to continuously receive updates from the server while issuing commands via the command line to view the game state or make a move
- Implemented shared memory to allow the server to spawn threads for each user which efficiently manages user requests, messages, and state updates
- **Available at:** <https://gitfront.io/r/ricwp12/U24fPLcj6QfM/Battleship/>

TECHNICAL SKILLS

Languages: Python, Java, C++, C, JavaScript, Rust, TypeScript

Databases: MySQL, MongoDB

Frameworks/Tools: Node.js, Express.js, React, Springboot

Other: Linux, Docker, Google Cloud Platform, Git, AWS